

Assignment of Chapter Two (B)

1). Fill-in-the-blank Questions:

- 1 $F = A \odot B$, its Minterm expression is _____. its Maxterm expression is _____.
 - 2 A function with n variables can form _____(how many) Maxterms, _____(how many) Minterms.
 - 3 A canonical "AND-OR" expression is made of _____.
 - 4 $F(A, B, C) = \sum m(1, 2, 3, 4, 7)$ can be converted to another canonical form, whose abbreviated expression is _____.
 - 5 If $F(A, B, C) = \prod M(0, 2, 5)$, then $F'(A, B, C) =$ _____.
 - 6 The universal gates usually refer to _____ and _____, such two logic gates.
- 2). List the "SOM" and "POM" of the following three logic functions (**Do not use the shorthand form**)
- 1, $F(A, B) = \sum m(1, 3)$

2, $F(A, B, C) = \sum m(1, 4, 6, 7)$

$$3, \quad F(A, B, C) = \prod M(0, 1, 2, 4, 5)$$

$$4, \quad F(A, B, C, D) = \overline{B}\overline{C}\overline{D} + \overline{A}B + A\overline{B}\overline{C}D + BC$$

SOM

$$5. \quad F(A, B, C, D) = \overline{(\overline{A}\overline{B} + ABD)} + (B + CD)$$

SOM